

Information-Preserving Quantization of Domain-Specialist Fine-Tunes for Verification in Multi-Agent Scientific Reasoning Systems

Research Proposal

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Abstract

Multi-agent large language model (LLM) systems improve reasoning, factuality, and domain coverage over single-model baselines, but at substantial cost — running multiple frontier models per query is expensive and excludes researchers without API budgets. We propose a hybrid architecture in which frontier API models serve as primary reasoners while locally-hosted, fine-tuned, aggressively-quantized open-weights models act as a *verification layer*, auditing domain-specific claims that frontier voices make. We hypothesize that aggressive quantization of a small domain-specialist fine-tune — when used for verification rather than open-ended generation — preserves enough domain-relevant knowledge to detect specific failure modes that frontier models systematically miss. Using Gemma-4-31B fine-tuned via LoRA on a gravitational-wave (GW) and signal-processing corpus, we will conduct a controlled ablation across six bit-depths (FP16, Q8, Q6, Q4, Q3, Q2) comparing post-training quantization (PTQ) and quantization-aware training (QAT). The contribution is threefold: (i) a system architecture for cost-efficient hybrid frontier-local scientific debate; (ii) a quantization-degradation curve identifying the practical floor of usable specialist-as-verifier capability; (iii) a comparative study of PTQ vs. QAT in the specialist-verifier regime.

1 Background and Motivation

1.1 The cost-quality bottleneck in multi-agent scientific AI

Multi-agent LLM systems are now a leading paradigm for AI-assisted scientific reasoning. Google’s AI Co-Scientist [1] — six Gemini-2.0 agents in a closed-loop tournament — demonstrated novel hypotheses in biomedical discovery. The “Council” and `/discover` modes of the author’s prior engineering work implement debate-style multi-agent reasoning [2–4] for code review and research-question deliberation. Across this paradigm, *the per-query cost is dominated by frontier-model API calls*: each voice in an N -voice debate requires its own frontier inference.

This cost structure has two consequences. First, it excludes researchers without API budgets — a single `/discover` run can cost \$1–3 across seven frontier API calls. Second, it penalizes high-volume inner-loop operations: in Co-Scientist’s tournament structure, Reflection and Ranking agents run dozens of times per cycle. Aggressive cost reduction would unlock single-graduate-student-budget access to scientific multi-agent systems — a meaningful democratization of the methodology.

1.2 Why local quantized models are not (currently) drop-in replacements

The natural alternative — replacing frontier models with local quantized models — fails empirically. Frontier models on benchmarks like GPQA [5] and MMLU-Pro [6] outperform 30B-class open-weights models by 15–30% absolute on graduate-level scientific reasoning. The capability gap is real and disproportionately affects the *generation* and *novel reasoning* roles where breadth matters most.

Recent work by Liu et al. [7] showed that quantization disproportionately damages mathematical reasoning compared to general language ability: AWQ [8] and GPTQ [9] introduce up to 32.39% accuracy degradation on math benchmarks, while showing only mild impact on commonsense tasks. *Quantization is not a free cost reduction for reasoning-heavy work.*

1.3 The architectural insight

Frontier models and quantized specialists are not in direct competition; they have complementary failure modes:

- **Frontier models** reason broadly but confidently hallucinate domain specifics — including arithmetic slips, wrong scaling laws, and misremembered constants. The author has directly observed these failure modes in `/discover` runs over the GW-quantization research domain.
- **Quantized domain specialists** have narrower knowledge but can be tuned to *flag* domain-specific errors without needing to *generate* novel content.

This suggests a hybrid: frontier reasoners produce candidate positions; a quantized specialist *verifies* their domain-specific claims. The specialist is not asked to do what it cannot (reason novelty); it is asked only to recognize what it has been trained on (correctness of specific claims in its domain).

1.4 The quantization research opportunity

A specialist’s value to the system depends on how much capability survives quantization. Liu et al. [7] showed that small-scale fine-tuning (~545 examples) can restore much of the quantization damage on a narrow reasoning task. *What is unknown* is how that finding extends to the specialist-as-verifier setting in a multi-agent context — where the specialist is asked structured claim-evaluation questions rather than open-ended math problems.

If the quantization-degradation curve has a knee at Q4 or below, the system architecture becomes practical for any researcher with consumer-grade hardware. If the curve is monotonic-with-no-knee, the result is sobering but informative for the field. **Either result is publishable.**

2 Research Questions

The project addresses four nested questions in decreasing scope.

RQ1 (system-level, primary)

Does augmenting a frontier-model multi-agent reasoning system with a quantized domain-specialist verification layer improve domain-specific output quality at a meaningful improvement in cost–quality Pareto frontier?

Operationalization: measure absolute and relative improvement in eval rubric scores (5-dimensional, 0–3 each) on a held-out set of GW-domain research questions, comparing (a) frontier-only, (b) frontier + specialist FP16, (c) frontier + specialist at each of Q8/Q6/Q4/Q3/Q2, (d) all of the above + MCP grounding.

RQ2 (quantization curve, headline contribution)

What is the shape of the quality-vs-bit-depth degradation curve for a domain-specialist verifier, and at what bit-depth does the specialist stop being useful?

Operationalization: measure eval rubric scores per bit-depth, identify discontinuities (“knees”), characterize *which classes of errors* the specialist starts missing at each bit-depth.

RQ3 (PTQ vs. QAT)

Does quantization-aware training [10] extend the useful bit-depth range beyond post-training quantization by a meaningful margin in the specialist-verifier regime?

Operationalization: at four critical bit-depths (Q8, Q4, Q3, Q2), train QAT variants alongside PTQ variants. Quantify the bit-depth headroom QAT provides.

RQ4 (interaction with tool use)

How does augmenting the verification layer with external tool use (arXiv MCP, Wolfram MCP) interact with specialist quantization — is it additive, multiplicative, or substitutive?

Operationalization: $2 \times 2 \times 6$ ablation (specialist on/off $\times$ MCP on/off $\times$ 6 bit-depths); per-failure-class analysis of where each layer catches errors.

3 Hypotheses (Pre-Registered)

We commit to the following predictions to be evaluated against Phase 3–4 results:

- H1** Adding an un-quantized specialist (FP16) to the frontier-only baseline improves GW-domain eval scores by $\geq 10\%$ absolute.
- H2** The quantization-degradation curve has a knee between Q4 and Q3 (Q8–Q4 roughly equivalent; Q3 and below show steep decline).
- H3** QAT extends the usable bit-depth range by approximately one bit-depth versus PTQ (QAT-Q3 $\approx$ PTQ-Q4 in usefulness).
- H4** MCP augmentation and specialist verification are *complementary* (different error classes), so frontier+MCP+specialist > frontier+MCP > frontier+specialist > frontier.

Null-results explicitly accepted: H1 failure would falsify the entire specialist-verifier concept — itself a publishable finding. H2 failure (monotonic degradation, no knee) is the most likely *boring* outcome but still publishable as a characterization of practical limits. H3 failure (QAT = PTQ) would be an important negative result for QAT advocates in the specialist regime.

4 Methodology

4.1 Base model and fine-tuning

Base: Gemma-4-31B [11] (or fallback Gemma-3-27B if 31B is not actually available — methodology is base-agnostic).

Fine-tuning: LoRA [12] via HuggingFace PEFT (or Unsloth for speed). Initial hyperparameters: rank 64, alpha 16, learning rate 1×10^{-4} , batch size 4 with gradient accumulation 4, target modules {q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj}. Hyperparameters

refined in Phase 2 via held-out eval. We will also benchmark DoRA [13] against LoRA as a candidate improvement.

Corpus for the GW-quantization specialist:

- arXiv abstracts from `gr-qc`, `astro-ph`.IM, `eess.SP` (last 5 years, ~ 30 K abstracts; full text where OA permits)
- Author’s existing literature review on GW + quantization SNR degradation
- Author’s unpublished derivations and notes (highest per-token value because not in any frontier training set)
- Selected textbook sections (Maggiore; Creighton & Anderson)
- General instruction data (FLAN/Alpaca, 20–30% mix) to mitigate catastrophic forgetting [14]

Target corpus size: 50–150M tokens after deduplication. Phase 2 will validate this is sufficient via held-out perplexity.

4.2 Quantization protocols

Post-Training Quantization (PTQ): GGUF formats via llama.cpp [15]: FP16 (reference), Q8_0, Q6_K, Q4_K_M, Q3_K_M, Q2_K.

Quantization-Aware Training (QAT): LLM-QAT methodology [10] — synthetic data distillation from the FP16-fine-tuned teacher; quantize during training; weights, activations, and KV cache. Train at four critical bit-depths (Q8, Q4, Q3, Q2) — skip intermediates for time.

4.3 Verification-layer architecture

The verifier operates on extracted claims, not whole positions:

1. Frontier voice produces a position.
2. A *claim extractor* (lightweight frontier-model call) parses the position into a JSON list of quantitative or factual claims.
3. For each claim, the specialist outputs $\{\text{verdict} \in \{\text{CONSISTENT}, \text{INCONSISTENT}, \text{UNCERTAIN}\}, \text{reasoning}, \text{suggested_probe} \in \{\text{arxiv}, \text{wolfram}, \text{code}, \text{none}\}\}$.
4. Suggested probes are dispatched to MCPs [16] (Phase 4).
5. Annotated position is forwarded to the Synthesist for final brief generation.

This deliberately keeps the specialist’s task *structured and bounded*. The hypothesis is that a small quantized model can do “evaluate this claim against my domain training” even when it cannot do “generate a novel hypothesis.”

4.4 Evaluation

Primary eval set: 25 hand-curated research-question briefs spanning GW physics (in-domain), general physics (adjacent), CS/ML (out-of-domain). Manually scored against a 5-dimensional rubric (factual correctness, math correctness, citation quality, novelty, structural completeness), each scored 0–3. Rubric reliability validated via blind re-grade at 7-day interval (target Cohen’s $\kappa \geq 0.6$) and peer blind-grade on 5-item subset.

Secondary eval set: MMLU-Pro [6] physics + chemistry subsets (10 questions each) as the out-of-domain reference, providing a general-reasoning anchor independent of the hand-curated GW set.

Per-cell sample size: 25 (in-domain) + 20 (MMLU-Pro). Bootstrap confidence intervals reported for all key comparisons. Rubric scoring blind to configuration; order randomized.

4.5 Ablation matrix

Full factorial: 2 (specialist on/off) $\times$ 6 (bit-depth, including FP16) $\times$ 2 (MCP on/off) $\times$ 2 (domain). In practice, ~ 30 cells filled (some combinations uninformative). Each cell $\times \sim 25$ evals = ~ 750 total grading operations, manageable within the 7-week timeline.

4.6 Statistical analysis

Per-cell: mean rubric score per dimension, with bootstrap 95% CIs. Pairwise comparisons: paired comparison (same eval items) between cells; report effect size + p -value with Bonferroni-corrected $\alpha = 0.05/k$ for k comparisons. Quantization curves: scores plotted vs. effective bits per parameter, with shaded CI band.

4.7 Reproducibility

All artifacts will be open-sourced: LoRA adapter weights (HuggingFace Hub), quantized GGUF variants (HuggingFace Hub), eval set + rubric + scoring scripts (GitHub), training and inference scripts (GitHub), frozen Python environment (`requirements-lock.txt`), and compute logs (GPU hours, peak memory).

5 Expected Contributions

5.1 Empirical

- First controlled quantization sweep of a domain-specialist verifier in a multi-agent context (across 6 bit-depths, PTQ + QAT, both in-domain and out-of-domain evaluation).
- Cost-quality Pareto frontier for hybrid frontier + local-specialist systems on scientific reasoning tasks.
- Per-error-class degradation profile: which kinds of errors the specialist starts missing at each bit-depth.

5.2 Methodological

- Reproducible eval harness for multi-agent scientific reasoning systems, with hand-graded rubric methodology + reliability validation.
- Claim-extractor + verifier interface design for plugging local specialists into existing multi-agent frameworks.

5.3 System / engineering

- Open-source verification layer implementation for Council and /discover.
- MCP-augmented verifier wiring (arXiv + Wolfram).
- Per-domain specialist artifacts (starting with GW, extensible).

5.4 Connections to the author’s existing research

The project directly extends the author’s prior work on quantization-induced SNR degradation in GW trigger pipelines from *detector signal quantization* to *model weight quantization*, using a unified rate-distortion framework. This makes the project a coherent thesis-portfolio piece rather than a one-off engineering exercise.

6 Evaluation Plan and Decision Gates

Phase	Decision gate	Pass	Fail action
0	Rubric reliability	Cohen’s $\kappa \geq 0.6$	Extend 1 week; commit even if imperfect
1	Failure-mode hypothesis	≥ 1 class addressable by specialist	Pivot to MCP-only architecture
2	FP16 specialist viability	$\geq 10\%$ absolute improvement on GW eval	Diagnose; if no fix, pivot to MCP-only
3	Quantization curve shape	Knee identifiable OR clear monotone curve	None — either is publishable
4	MCP $\times$ specialist interaction	Measurable effect (positive or negative)	None — descriptive only

Table 1: Per-phase decision gates and contingencies. The project has multiple paths to publishable outcomes; the only true failure mode is failing to produce any of them — which the Phase 0–1 design specifically guards against.

7 Timeline

Week	Phase	Milestone	Output
0	Prep	Environment ready	Locked deps, GPU verified, venue picked
1	P0	Eval harness locked	Rubric v1, 25 briefs, κ check
2	P1	Baseline characterized	Failure-mode taxonomy
3	P2.1	Specialist data prepared	Training corpus + config
4	P2.2	FP16 specialist validated	LoRA + Phase 2 writeup
5	P3.1	PTQ sweep complete	6 quantized variants + curve
6	P3.2	QAT sweep + comparison	Headline figure + paper draft v1
7	P4	MCP integration	Final ablation + paper draft v2
8	Submit	Paper submitted	Workshop submission

Table 2: Eight-week timeline from Week 0 prep to workshop submission. Detailed weekly todos in the companion `WEEKLY_PLAN.md`.

8 Resources Required

Compute

1 $\times$ A100 80GB rental (RunPod / vast.ai / Modal): ~ 60 – 100 GPU-hours across all phases. Estimated cost \$100–250. Local RTX 4090 (24GB) for eval and FP16/Q4 inference: ~ 50 GPU-hours, free.

API budget

Claude (Opus 4.7, Sonnet 4.6) + Gemini (3.5 Pro, Flash): $\sim \$100$ – 150 across all phases. Within existing subscription tiers.

Storage

~ 150 GB for model variants + corpus + eval. Already available.

Human time

~135–200 focused hours over 8 weeks: ~30% eval design + grading, ~30% training + experiments, ~40% writing.

Tooling

Ollama, HuggingFace `transformers` + `peft` (or Unsloth), `llama.cpp` for GGUF, `vllm` for serving, `matplotlib` for figures.

9 Risks and Mitigations

Risk	Likelihood Impact		Mitigation
Phase 0 rubric never reaches $\kappa \geq 0.6$	medium	blocks downstream	hard time-box at 2 weeks; commit if imperfect; document limitation
Specialist FP16 fails Gate 2 (H1 false)	medium-low	invalidates main hypothesis	diagnose (data quantity, base model, prompt design); if irreducible, paper becomes negative result
Quantization curve has no knee	medium	weaker paper	reframe as “limits of small-model specialization under PTQ”; still publishable
Gemma 4 31B does not actually exist	low	switch base model	methodology is base-agnostic; re-run with Gemma 3 27B or Qwen 2.5 32B [17]
Local GPU insufficient	medium	inference must be rented	Q4_K_M of 31B fits in 24GB; only training needs rental, budgeted
MCP layer breaks fault tolerance	medium	Phase 4 incomplete	benchmark MCP latencies before integration; raise timeouts; cache aggressively
Catastrophic forgetting in fine-tune	medium	specialist loses general reasoning	mix 20–30% general instruction data; validate against MMLU-Pro subset each epoch [14]
Three months of infra, no experiments	medium-high	highest risk	time-box hard at every phase; weekly writeups force discipline
Sample size insufficient for significance	medium	inconclusive results	$n = 25 + n = 20$; bootstrap CIs to detect noise; report inconclusive findings honestly
Workshop deadline missed	low	delayed publication	flexibility across multiple targets (NeurIPS ENLSP, ICLR ME-FoMo, ICML AI4Science); blog/preprint always available
Goodhart on the rubric	low	inflated results	held-out subset never used during prompt iteration; results on held-out only

10 Ethics, Open Science, and Reproducibility Commitments

- **No human-subject data.** All training corpus is public arXiv + open-access textbooks + author’s own notes.

- **No deceptive deployment.** The system is a research artifact; no claims will be made about safety for clinical, financial, or other high-stakes uses.
- **Open release** of all artifacts: code, weights, eval set, scoring scripts, hyperparameter logs.
- **Reporting of negative results:** if Phase 2 fails Gate 2, the failure mode and diagnostic process will be documented and submitted as a negative-results paper.
- **Pre-registration:** the four hypotheses in Section 3 are pre-registered. Any post-hoc hypotheses introduced after seeing results will be clearly labeled as exploratory.
- **Compute disclosure:** all GPU hours and API costs reported in the paper’s reproducibility section.

11 Beyond the First Paper

If Phases 2–4 succeed, three natural follow-ups become available:

1. **Multi-specialist scaling study:** does the quantization curve generalize across domains, or is it domain-specific?
2. **Closed-loop Co-Scientist with quantized verifiers:** integrate the verification layer into the full 6-agent hypothesis-tournament architecture inspired by [1].
3. **Quantization-aware specialization methodology:** develop new training recipes that produce specialists *designed* to survive aggressive quantization, rather than first training FP16 then degrading.

Each follow-up could be a self-contained paper. The present proposal commits only to the first.

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